© 2019 Human Kinetics, Inc.

Note: This article will be published in a forthcoming issue of the *International Journal of Sport Nutrition and Exercise Metabolism*. This article appears here in its accepted, peer-reviewed form; it has not been copyedited, proofed, or formatted by the publisher.

Section: Original Research

Article Title: Development of an Athlete Diet Index for Rapid Dietary Assessment of Athletes

Authors: Louise Capling[1,8], Janelle A. Gifford[1], Kathryn L. Beck[2], Victoria M. Flood[1,3,4], Gary J. Slater[5,6], Gareth S. Denyer[7], and Helen T. O'Connor[1,3]

Affiliations: [1]The University of Sydney, Discipline of Exercise and Sport Science, Faculty of Health Sciences, Lidcombe, NSW, Australia. [2]School of Sport Exercise and Nutrition, College of Health, Massey University, Auckland, New Zealand. [3]The University of Sydney, Charles Perkins Centre, Camperdown, NSW, Australia. [4]Western Sydney Local Health District, Westmead, NSW, Australia. [5]School of Health and Sport Sciences, University of the Sunshine Coast, Maroochydore, QLD, Australia. [6]Australian Institute of Sport, Canberra, ACT, Australia. [7]The University of Sydney, School of Life and Environmental Sciences, Camperdown, NSW, Australia. [8]Queensland Academy of Sport, Sport Performance Innovation and Knowledge Excellence, Brisbane, QLD, Australia.

Running Head: Development of an Athlete Diet Index

Journal: *International Journal of Sport Nutrition and Exercise Metabolism*

Acceptance Date: May 15, 2019

©2019 Human Kinetics, Inc.

DOI: https://doi.org/10.1123/ijsnem.2019-0098

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

"Development of an Athlete Diet Index for Rapid Dietary Assessment of Athletes" by Capling L et al.
International Journal of Sport Nutrition and Exercise Metabolism
© 2019 Human Kinetics, Inc.

Original research

Development of an Athlete Diet Index for rapid dietary assessment of athletes

Authors: Louise Capling[1,8*], Janelle A. Gifford[1], Kathryn L. Beck[2], Victoria M. Flood[1,3,4], Gary J. Slater[5,6], Gareth S. Denyer[7], and Helen T. O'Connor[1,3]

Affiliations:

1. The University of Sydney, Discipline of Exercise and Sport Science, Faculty of Health Sciences, Lidcombe, NSW, Australia.
2. School of Sport Exercise and Nutrition, College of Health, Massey University, Auckland, New Zealand.
3. The University of Sydney, Charles Perkins Centre, Camperdown, NSW, Australia.
4. Western Sydney Local Health District, Westmead, NSW, Australia.
5. School of Health and Sport Sciences, University of the Sunshine Coast, Maroochydore, Qld., Australia.
6. Australian Institute of Sport, Canberra, ACT, Australia.
7. The University of Sydney, School of Life and Environmental Sciences, Camperdown, NSW, Australia.
8. Queensland Academy of Sport, Sport Performance Innovation and Knowledge Excellence, Brisbane, Qld., Australia.

***Correspondence**: Louise Capling acap7726@uni.sydney.edu.au
Janelle A. Gifford (janelle.gifford@sydney.edu.au)
Kathryn L. Beck (k.l.beck@massey.ac.nz)
Victoria M. Flood (vicki.flood@sydney.edu.au)
Gary J. Slater (gslater@usc.edu.au)
Gareth Denyer (gareth.denyer@sydney.edu.au)
Helen T. O'Connor (helen.oconnor@sydney.edu.au)

Running title: Development of an Athlete Diet Index

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Abstract

Food-based diet indices provide a practical, rapid and inexpensive way of evaluating dietary intake. Rather than nutrients, diet indices assess intake of whole foods and dietary patterns, and compare these to nutrition guidelines. An athlete-specific diet index would offer an efficient and practical way to assess the quality of athletes' diets, guide nutrition interventions and focus sport nutrition support. This study describes the development and validation of an Athlete Diet Index (ADI). Item development was informed by a review of existing diet indices, relevant literature and in-depth focus groups with 20 sports nutritionists (median 11 years' professional experience) from four elite athlete sporting institutes. Focus group data were analyzed (NVivo 11 Pro) and key themes identified to guide development of athlete-relevant items. A modified Delphi survey in a sub-group of sports nutritionists (*n*=9) supported item content validation. Pilot testing with athletes (*n*=15) subsequently informed face validity. The final ADI (*n*=68 items) was categorized into three sections. Section A (*n*=45 items) evaluated usual intake; special diets or intolerances; dietary habits and culinary skills. Section B (*n*=15 items) assessed training load; nutrition supporting training; and sports supplement use. Section C (*n*=8 items) captured demographic details; sporting type and calibre. All athletes reported the ADI as 'easy' (40%) or 'very easy' (60% of participants) to use, and rated the tool as 'relevant' (37%) or 'very relevant' (63% of participants) to athletes. Further evaluation of the ADI, including the development of a scoring matrix and validation compared to established dietary methodology, is warranted.

Keywords energy intake, diet quality, sports nutrition

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Introduction

Dietary assessment in athletes is challenging for several reasons including the burden and difficulty of accurately reporting all foods and beverages consumed (Magkos and Yannakoulia, 2003), the expense and resources required to undertake dietary analysis and provide athlete feedback (Braakhuis et al, 2003), and the workload when large groups of athletes require assessment. For these reasons, dietary assessment may be conducted infrequently or not at all, making it difficult for sports nutritionists to identify athletes requiring dietary support or to evaluate the effectiveness of nutrition interventions. Additionally, diet assessment methods have had limited validation in athletes (Capling et al., 2017). Typically methods suitable for the general population (e.g. food record, 24-hr dietary recall) are applied in a similar manner to athletes despite known athlete-specific issues such as a variable energy intake, periodized energy expenditure, frequent eating occasions, and use of sports foods and dietary supplements which complicate accurate dietary assessment (Magkos and Yannakoulia, 2003). As a poor quality diet may compromise recovery, adaptation from training or increase risk of illness or injury (Thomas et al., 2016); valid, cost-effective and logistically feasible ways of evaluating the dietary intake of athletes are warranted.

Increasingly, nutritional epidemiology has shifted focus from single nutrients towards evaluating the whole diet or diet quality (Hu, 2002). A 'diet quality index' or 'diet quality score' applies a scoring system that compares usual intake or dietary patterns against dietary guidelines or other pre-defined criteria; with a higher score being positively associated with a more favourable dietary intake (Kant, 1996; Waijers et al., 2007; Wirt and Collins, 2009). A recent review (Burggraf et al., 2018) identified 57 indices, or their variations, primarily based on three established

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

population diet indices (the Diet Quality Index, the Healthy Eating Index and the Mediterranean Diet Score), however an athlete-specific diet index does not exist.

In recent years, the diet quality of athletes has occasionally been reported using population diet indices (Jürgensen et al., 2015; Spronk et al., 2015; Burrows et al., 2016). While there are aspects of population diet indices relevant to athletes (e.g. achieving the recommended servings of core food groups), several sport-specific factors require consideration. Nutritional needs of athletes often encompass a higher or more varied energy requirement, an increased need for specific macro- or micronutrients (Burke, 2001; Manore and Thompson, 2015; Thomas et al., 2016), and specific timing of nutrient intake (Thomas et al., 2016), in addition to practical considerations around rigorous training routines (e.g. irregular meal times, suppressed appetite following exercise). While a high quality dietary intake is desirable to minimize risk of nutritional deficiencies and promote health; adherence to sport-specific recommendations to optimize adaptation from training and maximize performance outcomes is also desirable (Thomas et al., 2016).

Technology-based dietary assessment methods offer an appealing and cost-effective approach, particularly for screening or monitoring large groups of athletes. Electronic data capture can automate certain aspects of dietary assessment, reduce the time and cost of administration, and improve participant compliance by reducing the burden of recording (Khanna et al., 2010; Stumbo et al, 2013; Gemming et al., 2015; Boushey et al., 2016). An electronic athlete diet index that rapidly evaluates usual intake and dietary habits as well as adherence to sports nutrition recommendations would be valuable for sports nutritionists and researchers. Therefore, the purpose of this study was to develop and validate an electronic Athlete Diet Index (ADI).

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Methods

Study Design

A mixed methods approach was undertaken to develop an ADI over three phases. A previously developed dietary tool (Blair et al., 2019) provided the initial framework for the ADI. Item development was informed by a review of population diet indices, relevant sports nutrition literature and in-depth focus groups with sports nutritionists, specifically to generate athlete-focused items (Phase 1). A modified Delphi survey and subsequent detailed coding of focus group transcripts determined content validity of the draft items (Phase 2). Athletes then pilot tested the ADI to assess instrument readability and face validity (Phase 3). An overview of the process is provided in Figure 1. The study was approved by the University of Sydney Human Research Ethics Committee (protocol numbers: 2016/781, 2018/514). Participants provided written informed consent prior to commencement of each study phase.

Phase 1: Focus Group Discussions

Accredited sports nutritionists (ASNs) were purposively selected from national institutes of sport in Australia and New Zealand with three or more ASN consultants. ASNs were recruited based on their expertise working within elite sporting program(s) and were required to have accreditation with a national sports nutrition professional organization and a minimum of two years' experience. Potential participants were contacted by one researcher (LC) via email to the sporting institute nutrition lead with a formal invitation to participate.

Focus group discussions were directed by a semi-structured interview schedule developed by the research team (Table 1). Transition and key questions were used to inform desirable components of an ADI. Provision of stimulus to the ASNs included examples of online population diet indices (Collins et al., 2015; Hendrie et al. 2017), and items from a previously developed

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

dietary tool (Blair et al., 2019). Participants were prompted to expand and explain their responses until they had no further input. All focus groups were facilitated by a moderator (LC) and assistant moderator/observer (JG or HOC) who were experienced in qualitative research and kept notes during the sessions. These investigators (LC, JG, HOC) were also experienced ASNs.

Data Analysis

The focus groups were audio-recorded then transcribed *verbatim* by a professional transcription service (Way with Words Ltd., Chatswood, Australia) before being returned to participants for verification and correction (i.e. incorrect speaker reference n=53, clarity of wording and/or supplementary detail n=21, and spelling mistakes n=6). ADI items were drafted and subsequently confirmed by the Delphi process (Phase 2) and detailed coding of the corrected de-identified transcripts. Transcripts were coded by one researcher (LC) using the semi-structured interview schedule as a framework (Table 1). Themes and sub-themes were identified via content analysis and refined by reading and re-reading transcripts. The coding tree hierarchy was examined by the researchers (LC, JG, HOC) for redundancies and cross-checked within existing themes to ensure consistency. Data was analyzed using NVivo 11 Pro qualitative data analysis software (QSR International Pty. Ltd., 2017, Melbourne, Australia).

Phase 2: Modified Delphi Survey

The ASNs who participated in the focus groups were invited to review the draft items and respond to a modified Delphi survey. The modified Delphi method involved presentation of a pre-selected list of items in an initial round to gain consensus of opinion from respondents (Hasson et al., 2000). The survey instructed participants to rate each item from 1 to 4 according to perceived relevance (i.e. 1 = "not relevant" to 4 = "highly relevant"), and indicate whether the item should be "retained", "deleted", or "modified" with an option to provide comments. The initial round

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

contained all draft items, while a second round only presented items where a modification and/or addition had been suggested. Feedback following the second round was reviewed by the investigators before item consensus was reached.

Phase 3: Pilot Study

A convenience sample of athletes from an Australian sporting institute was recruited to participate in the pilot study specifically to assess item comprehension, identify any ambiguous items, and determine relevance to athletes. Athletes (≥16 years) who had competed at state level or higher in their chosen sport were eligible. A formal e-mail invitation for athlete participants was sent to the sporting program head coach and ASN for dissemination.

A pilot ADI was developed using FileMaker™ Pro 14 software program (FileMaker, Inc., 2015, Santa Clara, U.S.A.) and deployed online via a portable electronic tablet (e.g. Apple™ ipad mini). Participants completed the ADI in the presence of one researcher (LC) where athletes were encouraged to seek clarity on unclear items. Field notes and time to complete the ADI were documented. Following completion of the ADI, athletes were asked to rate perceived level of instrument understanding from 1 to 5 (i.e. 1="very difficult" to 5="very easy"), and perceived relevance to athletes (i.e. 1="not relevant" to 5="highly relevant") (Supplementary material 1) to assess instrument readability and face validity.

Results

Phase 1: Focus Groups

Twenty (*n*=20; 15 female) experienced ASNs (median 11 years'; range 2-20 years' professional experience) participated in the focus groups. Experience encompassed a range of athlete type (i.e. endurance, team, combat, aesthetic and/or skill-based sports) and calibre (i.e. developmental, national, international and/or professional athletes).

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Focus groups were conducted during November, 2016 at four sites including three Australian and one New Zealand elite athlete sporting institutes. The focus groups lasted until participants had no further input between 100-110 minutes. The main themes identified included relevant features and adaptations for athletes; perceived utility of the ADI; evaluation of dietary intake and scoring; and details regarding ADI format and presentation. Athlete-specific adaptations deemed relevant for consideration in the development of an ADI included consideration of the higher or more variable energy intake of athletes; displaying athlete-relevant food images and serve sizes; details regarding training load and injury status; information about nutrition and hydration practices around training; meal patterns and dietary habits (e.g. food preparation, cooking, eating out of home); adherence to special diets and intolerances; plus sports supplement use (Table 2).

The perceived utility of the ADI included the potential for rapidly screening large groups of athletes, identification of athletes for follow up (i.e. 'red flags'); providing immediate feedback to the athlete and ASN; prioritizing resources (i.e. time and money); and advocating for nutrition servicing. Themes relating to the evaluation of dietary intake, scoring and reporting were also identified (Table 3).

Additional themes (not tabulated) related to the preferred instrument length, specific wording and use of images. When reviewing the online population indices and Massey University dietary tool, the sports nutritionists highlighted the importance of tool brevity, displaying suitable food images and using simple language. All themes were considered in the development of the draft ADI.

The previously developed dietary tool (Blair et al., 2019) provided an initial framework for the ADI. Relevant literature including the *Australian Guide to Health Eating* (NHMRC, 2013) and

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

sports nutrition position papers (Maughan and Shirreffs, 2011; Thomas et al., 2016), in conjunction with thematic data generated from the focus groups, informed the development of draft ADI items. Draft items were subsequently categorized into three sections. Section A (i.e. Core Nutrition) included a range of question formats to evaluate usual intake (i.e. core food groups, discretionary foods and alcohol), dietary habits (i.e. eating out of home) and culinary skills (i.e. cooking and shopping skills), special diets, intolerances and food avoidance, plus injury status. Section B (i.e. Performance Nutrition) involved sports-specific questions regarding training load (i.e. frequency, duration and type of training), nutrition and hydration practices around training, and sports supplement use. Finally, Section C (i.e. Personal Information) included standard items that captured demographic details, plus sporting type and calibre (exemplar questions from each section of the ADI are outlined in Supplementary material 2).

Phase 2: Modified Delphi Survey

Sports nutritionists (*n*=20) who participated in the focus groups were invited to participate in Phase 2. Four (*n*=4) declined due to lack of time or availability. Seven (*n*=7) participants did not respond to initial or follow up emails. Nine (*n*=9) agreed to participate and were presented with the draft items (*n*=71) between April and June, 2018. Of these, six items were recommended for deletion. Of the remainder (*n*=65), minor modifications were suggested for 54 items including revision of wording, and/or addition of sport specific text, and/or suggested images (Table 4). Approximately 72% (*n*=47 items) were perceived as "relevant" to "very relevant" by all participants. Two participants or fewer rated the remaining items (*n*=18) as "somewhat relevant" or "not relevant". Feedback was reviewed by the investigators (LC, HOC) and alterations made accordingly. The second round of the survey only presented modified (*n*=15) or additional items (*n*=3) and achieved consensus for inclusion by all respondents.

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Phase 3: Pilot Study

The pilot electronic ADI was developed between June-October 2018 and consisted of 68 items across the three sections: Section A (*n*=45 items); Section B (*n*=15 items); and Section C (*n*=8 items). Fifteen athletes were recruited to participate in Phase 3 and were presented with the pilot ADI between October and December, 2018. Demographic details of the participants are outlined in Table 5.

All participants rated perceived comprehension as "easy" (40% participants) or "very easy" (60% participants). All participants also rated the ADI as "relevant" (37% participants) or "very relevant*"* (63% participants) to athletes. All items (*n*=68) achieved consensus for inclusion with only minor modifications suggested 15 items (i.e. Section A *n*=8; Section B *n*=6; and Section C *n*=1) to reduce ambiguity. Modifications included revision of wording for clarity and/or addition of sport specific text. Key issues with comprehension or item ambiguity were identified, reviewed and incorporated into a revised version of the ADI.

The pilot ADI was completed by the athletes in 17.5 minutes (median; range 13-26 minutes). Overall the athletes reported the ADI as simple to use and were engaged by the interactive platform as indicated by the following quotations:

> *"I really liked the visuals of the serve sizes. This was very helpful. I know I eat more than an average person and average serve size, but having the pictures helped me work out how much I eat."* (A5).
>
> *"It would be great to do this more often to check how I am going, especially if there was a score and then I could see the change over time."* (A3)

Discussion

The results from this study support the content and face validity of the ADI and its perceived relevance and utility for use by athletes. The final ADI included 68 items presented over three sections (i.e. Core Nutrition, Performance Nutrition, and Personal Information). The concept

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

of an ADI was positively received by both ASNs and athletes. Athlete-relevant components were considered in the development of the ADI including an option to select a larger number of servings from some core food groups (e.g. grains), understanding nutrition and hydration practices around training, meal patterns and dietary habits, and sports supplement use. Identification of current training load, injury status and anthropometry were other features considered important for an ADI, in particular to address the influence of training periodization on current dietary intake.

Dietary assessment of athletes is challenging due to the influence of periodized training and other sports-specific factors such as frequent eating occasions, large portion sizes and widespread use of sports foods and supplements (Burke et al., 2003; Magkos and Yannakoulia, 2003). Advances in technology may facilitate automation of certain aspects of dietary assessment, reduce cost and respondent burden (Gemming et al., 2015; Boushey et al., 2016), however existing online dietary applications typically focus on assessing macro and/or micronutrient intake alone and have often not been validated in an athlete population. The ASNs in the current study discussed the merit of an ADI to evaluate usual intake of whole foods and dietary habits (i.e. meal patterns, cooking and shopping skills) in addition to adherence to sports nutrition recommendations. They identified the utility in rapidly identifying athletes at risk or capturing a change in intake due to periodized training, attempted body composition manipulation, or a disrupted routine (i.e. period away from home due to travel or competition).

It has been widely reported that athletes fail to meet sports nutrition recommendations across a range of sports (Heaney et al., 2010; Julian-Almarcegui et al., 2013; Wardenaar et al., 2017; Jenner et al., 2018). Sports nutrition guidelines encompass sports-specific energy and/or macronutrient recommendations to support training, maximize recovery and optimize performance outcomes (Maughan and Shirreffs, 2011; Thomas et al., 2016). While the diet quality of the

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

athletes participating in the pilot study was not evaluated, research indicates basic nutrition practices (i.e. an adequate intake of core food groups) to support optimal health and well-being may also be lacking. Jürgensen et al. (2015) observed a low consumption of fruits, vegetables, whole grains, milk and dairy products by team sport athletes. Burrows et al. (2016) noted a lack of variety within key food groups such as fruits, vegetables and dairy in rugby players despite overall diet quality being rated as 'good'. In practice, ASNs routinely apply various assessment methods to evaluate dietary intake of athletes (Larson-Meyer et al., 2018). The value of assessing diet quality and dietary habits rather than nutrient intake alone, along with the widespread use of electronic platforms across sporting programs, presents an opportunity for this novel electronic instrument. However, while early results indicate the ADI as a less burdensome way to rapidly assess diet quality and therefore could prove valuable to use on a larger population of athletes or within a team, it should not replace detailed dietary assessment or individual athlete guidance provided by the sports nutrition practitioner.

One of the strengths of the current study was the comprehensive process undertaken to determine relevant features and athlete-specific adaptations required in the development of an ADI with input from ASNs and athletes. While the themes generated from focus groups were from a modest sample of ASNs, all were highly experienced. Furthermore, while only a small number of athletes from a limited range of sports participated in the pilot study, results and feedback were consistent among participants. The prior sports nutrition experience of the researchers (LC, JG, HOC) may have unintentionally biased participant responses, however, a mutual understanding of the challenges faced in elite sport (e.g. finite resource, limited time, engagement of coaches and/or athletes) may have improved rapport with the ASNs and greater ability to probe themes. Finally, although this study was conducted in Australia and New Zealand, it would be relatively

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

straightforward to follow the study framework to make culturally-specific modifications and adapt to an international setting.

Conclusion

The results of this study support the content and face validity of the ADI plus its relevance and utility for use by athletes. Further evaluation of the ADI, including development of a scoring matrix and validation compared to established dietary methodology, is warranted.

Novelty statement

The Athlete Diet Index (ADI) is the first athlete-specific diet index to be developed.

Practical application statement

Sports nutritionists reported that the ADI could facilitate:

- Rapid assessment of athlete dietary patterns
- Identification of athletes at risk of inadequate or sub-optimal dietary intake
- Targeted nutrition intervention and evaluation of nutrition education

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Acknowledgments

The authors would like to thank the sports nutrition professionals and athletes who generously gave up their time to be a part of this study. Research Assistant, Dr. Wen Li and PhD candidate, Ryan Tam, are also gratefully acknowledged for their technical expertise in developing the pilot electronic tool.

Authorship

All authors contributed to the study design. Data were collected, analyzed and interpreted by LC, JG and HOC. Technical insights into the development of the electronic instrument were provided by GD. All authors contributed to manuscript preparation and final approval.

Disclosure of funding statement

This study was supported by a research grant from the Queensland Academy of Sport's Sport Performance Innovation and Knowledge Excellence Unit, Brisbane, Australia.

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

References

Blair, R.P., O'Connor, H.T., von Hurst, P.R., Conlon, C.A. and Beck, K.L. (2019) The validity and reproducibility of a screening tool to assess athletes adherence to food and nutrition guidelines. *British Journal of Nutrition* (under review)

Braakhuis, A.J., Meredith, K., Cox, G.R., Hopkins, W.G. and Burke, L.M. (2003) Variability in estimation of self-reported dietary intake data from elite athletes resulting from coding by different sports dietitians. *Int. J. Sport Nutr. Exercise Metab.* **13**:152-65.

Boushey, C.J., Spoden, M., Zhu, F.M., Delp, E.J. and Kerr, D.A. (2016) New mobile methods for dietary assessment: Review of image-assisted and image-based dietary assessment methods. *Proc. Nutr. Soc*. **76**:283–294.

Burggraf, C., Teuber, R., Brosig, S. and Meier, T. (2018) Review of a priori dietary quality indices in relation to their construction criteria. *Nutrition Reviews* **27**:1-18. https://doi.org/10.1093/nutrit/nuy027

Burke, L.M. (2001) Energy needs of athletes. *Can. J. Appl. Physiol*. **26**:S202–S219.

Burke, L.M., Slater, G., Broad, E.M., Haukka, J., Modulon, S. and Hopkins, W.G. (2003) Eating patterns and meal frequency of elite Australian athletes. *Int. J. Sport Nutr. Exercise Metab*. **13**:521-538.

Burrows, T., Harries, S.K., Williams, R.L., Lum, C. and Callister, R. (2016) The diet quality of competitive adolescent male rugby union players with energy balance estimated using different physical activity coefficients. *Nutrients* **8(**9**)**: 548.

Capling, L., Beck, K.L., Gifford, J.A., Slater, G., Flood, V.M. and O'Connor, H. (2017) Validity of dietary assessment in athletes: A systematic review. *Nutrients* **9**(1313); doi:10.3390/nu9121313

Collins, C.E., Burrows, T.L., Rollo, M.E., Boggess, M.M., Watson, J.F., Guest, M., Duncanson, K., Pezdirc, K. and Hutchesson, M.J. (2015) The comparative validity and reproducibility of a diet quality index for adults: the Australian Recommended Food Score. *Nutrients* **7***(*2), 785-798.

Farajian, P., Kavouras, S.A., Yannakoulia, M. and Sidossis, L.S. (2004) Dietary intake and nutritional practices of elite Greek aquatic athletes. *Int. J. Sport Nutr. Exerc. Metab.* **14**:574–585.

Gemming, L., Utter, J. and Ni Mhurchu, C. (2015) Image-assisted dietary assessment: A systematic review of the evidence. *J. Acad. Nutr. Diet*.**115**:64–77.

Hasson, F., Keeney, S. and McKenna, H. (2000) Research guidelines for the Delphi survey technique. *J Adv. Nurs.* **32**(4):1008-1015.

Heaney, S., O'Connor, H., Gifford, J. and Naughton, G. (2010) Comparison of strategies for assessing nutritional adequacy in elite female athletes' dietary intake. *Int. J. Sport Nutr. Exerc. Metab.* **20**(3):245-256.

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Hendrie, G.A., Baird, D., Golley, R.K. and Noakes, M. (2017) The CSIRO Healthy Diet Score: an online survey to estimate compliance with the Australian Dietary Guidelines. *Nutrients* **9:**47.

Hu, F.B. (2002) Dietary pattern analysis: a new direction in nutritional epidemiology. *Curr. Opin. Lipidol*. **13**(1):3-9.

Jenner, S.L., Trakman, G., Coutts, A., Kempton, T., Ryan, S., Forsyth, A. and Belski, R. (2018) Dietary intake of professional Australian football athletes surrounding body composition assessment. *J. Int. Soc. Sports Nutr*. **15**:43. https://doi.org/10.1186/s12970-018-0248-5

Jonnalagadda, S.S., Benardot, D. and Dill, M.N. (2000) Assessment of under-reporting of energy intake by elite female gymnasts. *Int. J. Sport Nutr. Exerc. Metab.* **10**:315–325.

Julian-Almarcegui, C., Gómez-Cabello, A., González-Agüero, A., Olmedillas, H., Gómez-Bruton, A., Matute-Llorente, A., Casajús, J.A. and Vicente-Rodríguez, G. (2013) The nutritional status in adolescent Spanish cyclists. *Nutr. Hosp*. **28**:1184–1189.

Kant, A.K. (1996) Indexes of overall diet quality: A review. *J. Am. Diet. Assoc*. **96**:785-91.

Khanna, N., Boushey, C.J., Kerr, D., Okos, M., Ebert, D.S. and Delp, E.J. (2010) An overview of The Technology Assisted Dietary Assessment Project at Purdue University. In Proceedings of the IEEE International Symposium on Multimedia, Ningbo, China. pp. 290–295.

Larson-Meyer, D.E., Woolf, K. and Burke, L. (2018) Assessment of nutrient status in athletes and the need for supplementation. *Int. J. Sports Nutr. Exerc. Metab.* **28**(2):139-158. https://doi.org/10.1123/ijsnem.2017-0338

Magkos, F. and Yannakoulia, M. (2003) Methodology of dietary assessment in athletes: concepts and pitfalls. *Curr. Opin. Clin. Nutr. Metab. Care* **6**:539-49.

Manore, M.M. and Thompson, J.L. Energy requirements of the athlete: Assessment and evidence of energy efficiency (2015). In Burke, L. and Deakin, V. (Eds.), *Clinical Sports Nutrition* 5th ed. (pp. 114-139). North Ryde, Australia: McGraw-Hill.

Maughan, R.J. and Shirreffs, S.M. (2011) IOC Consensus Conference on Nutrition and Sport, 25-27 October 2010, International Olympic Committee, Lausanne, Switzerland. *J. Sports Sci.* **29**(Suppl. 1), S1. doi:10.1080/02640414.2011.619339

Mielgo-Ayuso, J., Zourdos, M.C., Calleja-González, J., Urdampilleta, A. and Ostojic, S.M. (2015) Dietary intake habits and controlled training on body composition and strength in elite female volleyball players during the season. *Appl. Physiol. Nutr. Metab*. **40**:827–834.

National Health and Medical Research Council (2013) *Australian Guide to Healthy Eating*, Canberra: National Health and Medical Research Council. Available from: https://www.eatforhealth.gov.au/

Philippou, E., Middleton, N., Pistos, C., Andreou, E., and Petrou, M. (2017) The impact of nutrition education on nutrition knowledge and adherence to the Mediterranean Diet in adolescent competitive swimmers. *J. Sci. Med. Sport.* **20**(4):328-332.

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Shriver, L.H., Betts, N.M. and Wollenberg, G. (2013) Dietary intakes and eating habits of college athletes: Are female college athletes following the current sports nutrition standards? *J. Am. Coll. Health Assoc.* **61**:10–16.

Spronk, I., Heaney, S.E., Prvan, T. and O'Connor, H.T. (2015) Relationship between general nutrition knowledge and dietary quality in elite athletes. *Int. J. Sports Nutr. Exerc. Metab*. **25(3)**: 243–251.

Stumbo, P.J. (2013) New technology in dietary assessment: A review of digital methods in improving food record accuracy. *Proc. Nutr. Soc*. **72**:70–76.

Thomas, D.T., Erdman, K.A. and Burke, L.M. (2016) American College of Sports Medicine Joint Position Statement. Nutrition and Athletic Performance. *Med. Sci. Sports Exerc.* **48**(3):543-568.

Waijers, P.M.C.M., Feskens, E.J.M. and Ocke, M.C. (2007) A critical review of predefined diet quality scores. *Br. J. Nutr.* **97**(2):219-231.

Wardenaar, F., Brinkmans, N., Ceelen, I., Van Rooji, B., Mensink, M., Witkamp, R. and De Vries, J. (2017A) Macronutrient intakes in 553 Dutch elite and sub-elite endurance, team, and strength athletes: Does intake differ between sport disciplines? *Nutrients*, **9**(119); doi: 10.3390/nu9020119

Wardenaar, F., Brinkmans, N., Ceelen, I., Van Booij, B., Mensink, M., Witkamp, R. and De Vries, J. (2017B) Micronutrient intakes in 553 Dutch elite and sub-elite athletes: Prevalence of low and high intakes in users and non-users of nutritional supplements. *Nutrients*, **9**(142); doi:10.3390/nu9020142

Wirt, A. and Collins, C.E. (2009) Diet quality – What is it and does it matter? *Public Health Nutr*. **2**:2473-92.

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

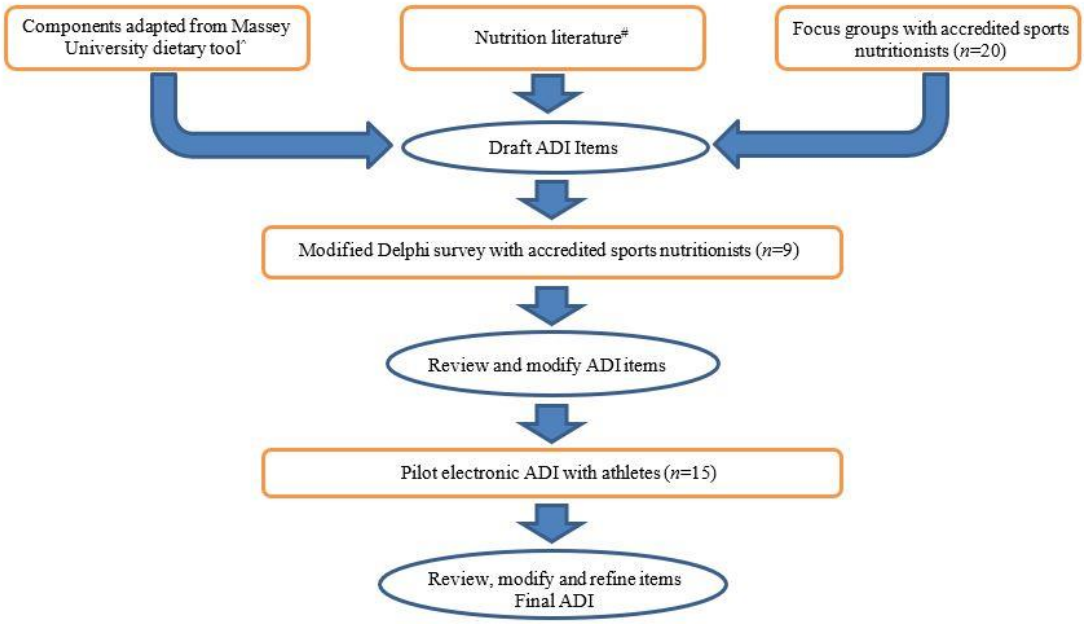

Figure 1: Methodological framework outlining the development and validation of an Athlete Diet Index

^ Components from a dietary tool previously developed at Massey University, New Zealand (Blair et al., 2019) provided the initial framework for the development of the ADI.

Relevant literature included a review of existing population diet indices, *Australian Guide to Healthy Eating* (NHMRC, 2013) and sports nutrition position statements (Maughan and Shirreffs, 2011; Thomas et al, 2016).

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Table 1: Interview schedule for focus group discussions

Question type	Questions
Opening	*Describe your professional experience working with elite athletes.*
Introduction	*What were some features you liked and disliked about the general population diet quality tools?*
Transition	*Would it be useful for sports nutrition professionals to assess the diet quality of athletes?*
	If a diet quality tool was available for athletes, what features would it include?
	What key elements do you think should be included in the dietary evaluation?
	What would the feedback look like for the athlete?
	Would the feedback be different for the sports nutrition professional?
Key	*What were some features you liked and disliked about the Massey University dietary tool?*
	What additional features do you think should be included specifically for athletes?
Ending	*What nutrition assessment methods do you use with your athletes?*
	Are you aware of other diet quality tools?
	Do you have any further suggestions for developing a diet quality tool for athletes?

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Table 2: Identified themes influencing athlete-specific adaptations and relevant features in the development of the ADI and exemplar quotations

Identified theme*	Quotation
Section A: Core Nutrition	
Energy requirements	"*What sport and maybe hours of training, or some question to get an idea of what their training looks like, because, obviously, a gymnast is going to need a whole lot less say, for instance, than a marathon runner for example*." G3-B
Core food groups	"*Maybe what we're sort of heading towards is maybe having two sections. So there's the healthy eating guidelines, then there's the sports nutrition section. So then we've got information on both*." G2-D
Frequency	"*People have a tendency to think about dinner especially where the protein serve is, but might forget about their lunch protein serve. So therefore the calculations are wrong if start doing it by the week or the month. I think the day is pretty accurate*." G2-B
Quantity	*"Standard serving size? They would triple or quadruple that for their standard serve size."* G2-B
Food images	"*Really helpful for athletes when they're answering questions to actually visualize what a serving is."* G3-C
Essential fats	"*Question on fat… I'd want it more from an inflammatory perspective*." G4-B
Alcohol	"*Most of them are not going to have one or two drinks, they are going to have either ten or zero*." G2-B

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Identified theme*	Quotation
Discretionary foods	"*… it could be quite difficult to generically talk about quality, because someone that's a weight category athlete ... you don't want them eating muesli bars, because it's too energy dense, but maybe a rower… four muesli bars … that's actually great*!" G3-C
Eating out/takeaway	"*The out of the home question is good because that takes into account breakfast. I think so many athletes eat breakfast out*. G4-D
Culinary skills	"*Could also include question about doing your own food shopping, cooking, or (determine if they live in) a family situation or (with a) partner*." G4-C
Special diets, food avoidance	*"(I would want to know) if they're avoiding certain foods*." G1-B
Influence on usual intake	"*Half the time they're either travelling or they're at a tournament so they're not eating anything like what they normally would and they're also training much less."* G3-C
Section B: Performance Nutrition	
Stage of training	"*Phase of training. So whether they're pre-comp(etition), post-comp(etition), off season, or injured*." G2-D
Training load	"*Their nutrition changes every day because of their training demands and the type of training*." G2-B

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Identified theme*	Quotation
Timing of intake	"*…breakfast, lunch and dinner, then some kind of question that looks into support of their training. I'm not saying you have to eat every time before your training and after your training, but often breakfast will be your recovery snack."* G3-C
Hydration	"*Do they drink before training? Do they routinely drink after training? Rather than an actual amount*." G4-A
Body composition	"*And (it depends) what their goals are. It could be that you're deliberately manipulating their carbohydrate at certainly meals of the day, you know, because you're trying to alter their body composition*." G4-A
Sports supplements	"*You can give the examples there. Because I think the big thing is athletes' perception of what a supplement is*." G1-C

*listed in order of item appearance in the ADI

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Table 3: Identified themes on the perceived utility of the ADI and evaluation of dietary intake and scoring with exemplar quotations

Identified theme	Quotation
Perceived utility of the ADI	
Advocacy role	"*If we have a tool that we all use, then potentially from a servicing point of view and saying to coaches, well this is the reason why I need to be doing more work with them because their score is actually really low*." G1-A
Prioritizing resources	"*Could be a good tool to prioritize. You're trying to disseminate nutrition resources or money then you can get 'thousands' of athletes to fill this in and figure out which need input more than others, and then divide the time and money."* G2-D
Practical application	*"Something like this could be used as a screening tool and could highlight to a whole lot of players who think are great but they're not, and who probably need some intervention."* G2-B
'Red flags'	"*This could be a tool that may alert people when they need to see you. Identification of red flags*." G1-B
Risk management	"*We don't do a lot of stuff on supplements because we're under-resourced, and it'll only take that one athlete to get banned for them (the coach) to go, oh hang on, we do need to invest in this area of servicing because it can happen so easily*." G3-A

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Identified theme	Quotation
Evaluation of dietary intake and scoring	
Evaluation summary	"*One (summary that's) athlete friendly and one more comprehensive that lists the score but then lists the flags for the practitioner*." G1-A
Relevant feedback	"*…you need to provide a reason why. If you eat (too many) discretionary foods, (it will) affect your body composition…focus on (the importance of) healthy fats from an anti-inflammatory point of view…eat more fruit to increase your nutrient density. Make it more sports-specific*." G3-C
Scoring	"*This athlete is likely to be not meeting their carbohydrate requirements, or they are less likely to be meeting their protein requirements because their diet quality score in this area is down."* G1-B
Nutrition follow up	"*Your carbohydrate intake is quite low, could suggest issues with recovery, fuelling. We'd advocate that you go and engage with your sports dietitian."* G1-C

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Table 4: Results generated from the Delphi survey with sports nutritionists categorized by section of the Athlete Diet Index

Responses from Delphi survey	**Section A** (*n*=48 items#)	**Section B** (*n*=16 items#)	**Section C** (*n*=7 items#)
Round 1 survey responses			
Number of items added	1		2
Number of items deleted	4	1	1
Number of items modified	35	12	7
Description of modifications			
Rephrasing of wording	21 mentions	6 mentions	7 mentions
Sports specific suggestions	3 mentions	8 mentions	3 mentions
Serve size or images	17 mentions		
*Round 2 survey presentation**			
Number of items modified	13	2	
Number of additional items	1		2

Total number of draft ADI items (*n*=71) presented in round 1 of Delphi survey. Section A Core Nutrition; Section B Performance Nutrition; Section C Personal Information.

* No further changes suggested following presentation of round 2 of Delphi survey

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Table 5: Mean (SD) demographics for age, self-reported stature (cm) and body mass (kg), and weekly training hours for athlete participants

Athlete *n*=15 (13F, 2M)	**Age** (years)	**Stature^** (cm)	**Body Mass^** (kg)	**Training** (hrs/week)	**Training Phase**	**Sport** (Representative level)
1 (F)	19	170	66	20	Pre-season	Water polo (National)
2 (F)	18	170	72	5	Injured	Water polo (International age)
3 (F)	19	173	72	16	Pre-season	Water polo (National)
4 (F)	18	174	70	11	Pre-season	Water polo (International age)
5 (F)	19	180	79	11	Pre-season	Water polo (National)
6 (F)	32	181	88	11	Pre-season	Water polo (International open)
7 (F)	17	171	68	20	In season	Rowing (International age)
8 (F)	22	179	74	20	In season	Rowing (International age)
9 (F)	20	182	71	20	In season	Rowing (International open)
10 (M)	18	188	91	20	In season	Rowing (International open)
11 (M)	18	194	88	20	In season	Rowing (International open)
12 (F)	17	183	72	20	In season	Rowing (International age)
13 (F)	18	174	69	20	In season	Rowing (International open)

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

"Development of an Athlete Diet Index for Rapid Dietary Assessment of Athletes" by Capling L et al.
International Journal of Sport Nutrition and Exercise Metabolism
© 2019 Human Kinetics, Inc.

Athlete *n*=15 (13F, 2M)	**Age** (years)	**Stature^** (cm)	**Body Mass^** (kg)	**Training** (hrs/week)	**Training Phase**	**Sport** (Representative level)
14 (F)	18	179	61	20	In season	Rowing (International age)
15 (F)	18	174	67	20	In season	Rowing (International age)
Mean ± SD	19.4 ± 3.7	178.8 ± 6.9	75.9 ± 8.8	16.9 ± 4.9		

^ Self-reported stature and body mass

M Male; F Female; SD standard deviation

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

"Development of an Athlete Diet Index for Rapid Dietary Assessment of Athletes" by Capling L et al.
International Journal of Sport Nutrition and Exercise Metabolism
© 2019 Human Kinetics, Inc.

Supplementary material 1: Evaluation survey for pilot study

1. ***Did you find the questions easy or difficult to understand?***

1	**2**	**3**	**4**	**5**
Very difficult	Somewhat difficult	Neutral	Somewhat easy	Very easy

2. ***Did you have any problems answering any questions? If so, what problems did you have?***

_____

3. ***Would you modify or delete any of the questions?***

_____

4. ***The final version of the Athlete Diet Index will provide a score and feedback to athletes.***

   How relevant do you think the Athlete Diet Index is for athletes?

1	**2**	**3**	**4**	**5**
Not relevant	Somewhat relevant	Neutral	Somewhat relevant	Very relevant

5. ***Do you have any other comments or suggestions for the Athlete Diet Index?***

_____

Time to complete ADI:_____mins

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

"Development of an Athlete Diet Index for Rapid Dietary Assessment of Athletes" by Capling L et al.
International Journal of Sport Nutrition and Exercise Metabolism
© 2019 Human Kinetics, Inc.

Supplementary material 2: Exemplar items from the Athlete Diet Index

Section A: Core Nutrition*	**Participant selection options**
Example items	
In the past week, how many servings of fruit (fresh, frozen, canned or dried fruit) did you usually eat each day?	None
	1 serve or less
	2 serves
Examples of one serving are equivalent to:	3 serves
(visual representation of standard fruit serves is provided)	4 serves or more
In the past week, how many days did you eat fruit?	None
	1 day
	2 days
	3 days
	4 days
	5 days or more

*Additional items assess the quantity and frequency of intake of core food groups (i.e. fruits, vegetables, grains, dairy, meat and alternatives), discretionary foods and alcohol. Other items explore dietary habits, culinary skills, special diets, intolerances and food avoidance.

© 2019 The University of Sydney and Massey University

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Section B: Performance Nutrition **Example items**	**Participant selection options**
Please indicate the current stage of your training cycle:	Pre-season In-season (training and/or competition) Off season (light training or rest) Injured (modified training or rehabilitation) Injured (no training, rest)
Are you currently trying to change your body composition? (e.g. increase muscle mass, decrease body fat, make weight for competition)	Yes No (If participant selects "Yes" then a new option is revealed):
Please specify:	Increase muscle mass Decrease body fat Make weight for competition Other

**Additional items explore current training load, nutrition and hydration practices supporting training, and sports supplement use.

© 2019 The University of Sydney and Massey University

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC

© 2019 Human Kinetics, Inc.

Section C: Personal Information* **Example items**	**Participant selection options**
What is your primary competitive sport? *(please select from list)*	(list of sports provided)
Has your weight changed in the past six months?	No change Gained 0-2 kg Gained 2-4 kg Gained 5kg or more Lost 0-2kg Lost 2-4kg Lost 5kg or more Unsure of weight change

******* Additional items capture demographic details such as sex, age, height, weight, and education level.

© 2019 The University of Sydney and Massey University

Authenticated t_lambrinides@yahoo.com | Downloaded 07/31/19 12:42 PM UTC